TITLE : A Behavior Consistency Framework for AI-Based Suspicious Transaction Detection in Banking

ABSTRACT

Financial fraud has become a major challenge for banks and digital payment systems due to the rapid growth of online transactions. Traditional fraud detection methods mainly rely on predefined rules and existing transaction features, making them less effective in identifying sophisticated fraudulent activities. This project proposes an AI-based financial fraud detection framework using the XGBoost machine learning algorithm to classify transactions as genuine or fraudulent. The baseline model is trained using the PaySim dataset and evaluated using Accuracy, Precision, Recall, F1-score, and Confusion Matrix. To overcome the limitations of existing approaches, the proposed framework introduces two novel behavioral features: Receiver Familiarity Score (RFS), which measures the historical familiarity between sender and receiver, and Behavior Consistency Score (BCS), which evaluates whether a transaction follows the customer's usual transaction pattern. These additional features provide contextual and behavioral information that helps reduce false negatives and improve fraud detection performance. The proposed framework aims to enhance the accuracy and reliability of banking fraud detection systems while minimizing financial losses.

INTRODUCTION

With the rapid growth of digital banking, mobile payments, and online financial services, the number of electronic transactions has increased significantly. While these technologies provide convenience and speed, they also create opportunities for fraudulent activities such as unauthorized fund transfers, identity theft, and money laundering. Detecting fraudulent transactions has therefore become a critical challenge for financial institutions.

Machine learning techniques have become widely adopted for fraud detection because they can automatically learn complex transaction patterns from historical data. Among these techniques, XGBoost is recognized for its high prediction accuracy and efficient handling of structured datasets. However, existing fraud detection models primarily rely on transaction-related features such as transaction amount, account balances, and transaction type. They often fail to capture the behavioral relationship between customers, leading to missed fraud cases and false alarms.

To address this limitation, this project proposes an enhanced AI-based fraud detection framework by introducing two behavioral features: Receiver Familiarity Score (RFS) and Behavior Consistency Score (BCS). Receiver Familiarity Score evaluates the historical interaction between the sender and receiver, while Behavior Consistency Score measures whether the current transaction is consistent with the customer's usual transaction behavior. These behavioral features are integrated with the existing transaction attributes and used to train an XGBoost model for improved fraud detection. The proposed framework is evaluated using the PaySim dataset and performance metrics such as Accuracy, Precision, Recall, F1-score, and Confusion Matrix to demonstrate its effectiveness over the baseline model.